Microstructure → hardness: a measurement-bias diagnosis

Image regression of a synthetic hardness proxy from 256×256 grayscale micrographs (500 train / 1000 test, RMSE). Protocol: one shared RepeatedKFold(5×3, seed 42) over images for every model, so all comparisons are paired by fold. No external data, no remote APIs; the only pretrained weights are timm ImageNet-1k checkpoints under Apache-2.0.

1. What the data is (and is not)

Labels are unrounded float64 generator output (100% of strings are shortest round-trip reprs); no value grid (Rayleigh $R \leq 0.09$ for every spacing); one PNG encoder signature for all files; no train/test shift (0/28 statistics significant by Bonferroni-corrected KS). A rotation/flip-invariant structure signature with a positive control of power 1.00 found no reused base structures; the 54 pairs flagged by the robust null have label gaps larger than random pairs (33.1 vs 19.7), i.e. no leak.

2. Central finding: the classic estimators measure the nuisance

Blur, noise (σ 4–20) and contrast vary strongly per image and are label-independent (|ρ| with hardness ≤ 0.04). Of all structural features, only 37/167 correlate more with hardness than with blur (Fig. 1). The ASTM E112 mean linear intercept has ρ_y = -0.027 but ρ_{noise} = +0.756; a Hall-Petch power law fitted on the estimated latents gives a grain-size exponent of +0.019 (physics: –0.5) and $R^2(\log y) = 0.22$. Three principled blur corrections — dividing the fitted Gaussian MTF out of the power spectrum, Richardson-Lucy deconvolution before re-segmentation, and scale-space extrapolation to zero blur — left 130 estimators blur-dominated; only 1 of 60 corrected features (a phase fraction, not a grain size) crossed the diagonal.

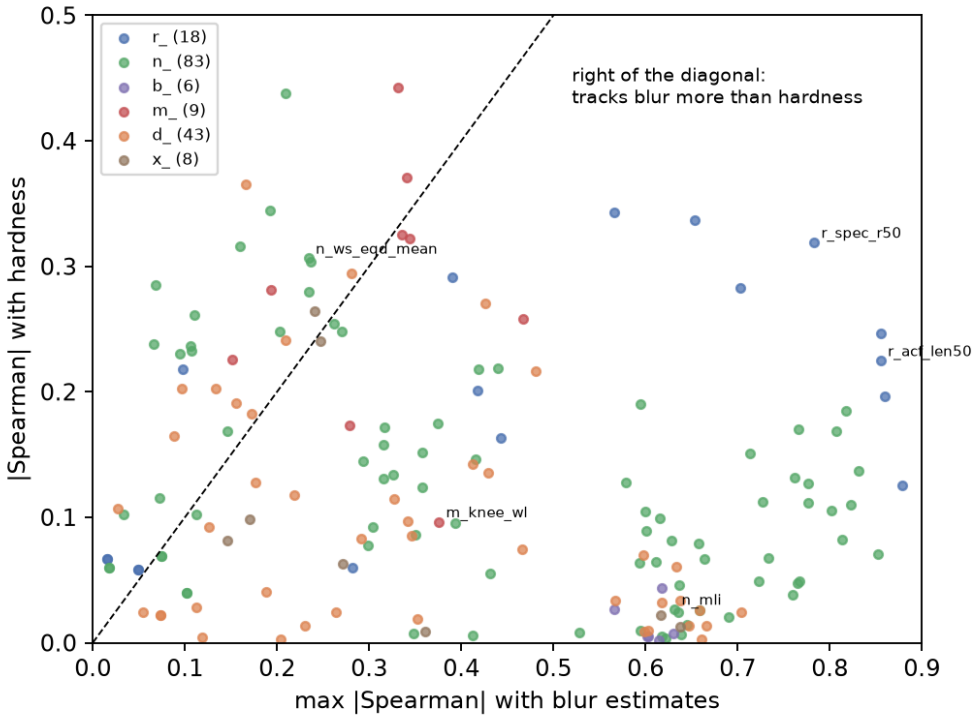


Fig. 1. Each point is one feature. Grain-size estimators (labelled) sit far below the diagonal before and after blur correction.

3. Ceiling analysis

Feature-space twins: the closest 1% of image pairs in the full feature space still differ by 14.7 in hardness against 19.7 for random pairs; restricting to non-blur-dominated features does not help (15.2). The best model's residual (sd 14.3) is uncorrelated with every feature (max |ρ| 0.11, none Bonferroni-significant) and unlearnable from them (CV R^2 -0.02). Quadrant ICC of the predictive structural features is only 0.60, so 128-px crops are not label-equivalent and the CNN trains on full images.

4. Models (RMSE, mean ± sd over 15 folds)

model	CV RMSE	R^2
constant mean	17.665 ± 1.329	-0.011
Ridge, 28 cheap global stats	14.853 ± 1.267	+0.281
LightGBM, cheap stats	14.993 ± 1.310	+0.269
LightGBM, 114 Stage-1 + cheap	14.568 ± 1.159	+0.308
LightGBM, + blur-corrected (m_/d_/x_)	14.514 ± 1.227	+0.313
best enumerated closed-form formula	15.531	–
blend eq(ridge_cheap+lgbm_s1)	14.354 ± 1.199	Δ vs 003 +0.000 (p 1)
blend nnls_cf(ridge_cheap+lgbm_s1)	14.353 ± 1.212	Δ vs 003 -0.000 (p 0.99)
blend eq(ridge_cheap+lgbm_s1+lgbm_s1new)	14.297 ± 1.202	Δ vs 003 -0.057 (p 0.13)
blend nnls_cf(ridge_cheap+lgbm_s1+lgbm_s1new)	14.330 ± 1.244	Δ vs 003 -0.024 (p 0.59)

Reproduction: `python src/run.py --mode full`. Environment, pinned versions and licenses: README.md. Every experiment with its CV number: experiments.md.